

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT**(25CS1302E)****2025-26, Term-III****Project Problem Statement Submission Form**

Sec No-	Team No-	Project Title-
Team Lead Roll Number		2520030524
Team Lead Name		Jawaji Advait
Team Member 1 Roll.No		2520030258
Team Member 1 Name		Bhimavarapu Sai Saranya
Team Member 2 Roll.No		2520030219
Team Member 2 Name		Challa Sreeja

1. Abstract

The **Society Management & Resident Services Platform with Distributed Backend Architecture** is a centralized web-based solution designed to simplify and automate the management of residential communities. The platform provides residents, society administrators, security personnel, and service providers with an efficient system for handling daily activities such as resident registration, visitor management, maintenance requests, announcements, facility bookings, complaints, and communication. The platform aims to replace manual registers and informal WhatsApp-group coordination with a structured digital system for maintenance billing, visitor tracking, complaint resolution, facility management, and society notices. A distributed backend architecture is used to improve scalability, reliability, performance, and secure data management by separating major functionalities into independent services. These services communicate through an API Gateway, providing better fault isolation and scalability compared to a single monolithic server.

The core technology stack consists of **React.js with Tailwind CSS** for the frontend, **Node.js with Express.js** for backend services, and **MySQL** as the relational database. **Redis** is used for caching frequently accessed data to improve system performance. The platform integrates database management and distributed backend technologies to provide a reliable, scalable, and efficient solution for modern residential community management.

2. Module Mapping

Module Name	Assigned Roll.No	Signature
Frontend Integration & Testing	2520030524	Jawaji Advait
API Development & Microservices	2520030258	Bhimavarapu Sai Saranya
Backend Architecture & Database Design	2520030219	Challa Sreeja

3. Module Details

Module	Responsibilities
Frontend Integration & Testing	Develop React.js interfaces; create resident, administrator, and security dashboards; integrate frontend with REST APIs; implement forms and client-side validation; perform API, functional, and integration testing; identify and fix defects; prepare the system for final demonstration.
API Development & Microservices	Develop RESTful APIs; implement Node.js/Express backend services; develop FastAPI API Gateway; implement JWT authentication and RBAC; design microservices and service boundaries; implement inter-service communication; handle validation, errors, and service-level integration.
Backend Architecture & Database Design	Design the overall system architecture; create ER diagrams and relational schemas; implement MySQL databases; apply normalization, constraints, indexing, views, stored procedures, triggers, and transactions; design database interactions for distributed services; manage database configuration and integration.

4. Frameworks / Platforms / Software Used

Category	Technology / Tool
Frontend	React.js
UI Framework	Tailwind CSS
Backend	Node.js + Express.js
API Gateway	FastAPI
Relational Database	MySQL
NoSQL Database	MongoDB
Caching	Redis
API Communication	REST APIs
Authentication	JWT
Database ORM / ODM	Sequelize / Mongoose
API Testing	Postman
Testing	Jest / Supertest
Containerization	Docker + Docker Compose
Version Control	Git + GitHub
API Documentation	OpenAPI / Swagger
Development Environment	Visual Studio Code

Guide Name and Signature (remarks if any) _____